Inch-Pound
MIL-DTL-25988D
21 January 2021
SUPERSEDING
MIL-DTL-25988C
11 August 2006

DETAIL SPECIFICATION

RUBBER, FLUOROSILICONE ELASTOMER, OIL-, HYDRAULIC FLUID-, AND FUEL-RESISTANT, SHEETS, STRIPS, MOLDED PARTS, AND EXTRUDED SHAPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Form. This specification covers a fluorosilicone (FVMQ) rubber in the form of molded rings, compression seals, O-ring cord, and molded-in-place gaskets and rubber sheets, strips, molded parts and extruded shapes for aeronautical and aerospace applications. This material is not suitable for use in synthetic phosphate ester based hydraulic fluids and generally not suitable for use in MIL-PRF-7808, MIL-PRF-23699 gas turbine lubricants or DOD-PRF-85734, MIL-PRF-32538 helicopter transmission lubricants. In future designs it is recommended to not use silicone sealing material in engine oil systems.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Wright-Patterson AFB OH 45433-7017 or emailed to engineering.standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.2 Application. This fluorosilicone (FVMQ) has resistance to hydraulic fluids, gas turbine engine oils, and fuels, but usage is not limited to such applications. This fluorosilicone (FVMQ) has a typical service temperature range of -70 to 392 °F (-56.7 to 200 °C) for Class 1 and Class 2 and -70 to 437 °F (-56.7 to 225 °C) for Class 3. The service temperature range of the material is a general temperature range, but the presence of particular fluids and specific design requirements may modify this range. Each application should be considered separately. It is the responsibility of the user to determine that this specification is appropriate for the environments (temperature range, fluid exposure, etc.) in which it is sought to be used.

Note: The evolution of synthetic ester engine oil formulations has resulted in the determination that fluorosilicone elastomers are no longer suitable for use in high thermal stability (HTS) lubricants. Therefore, this specification should not be used for new design with HTS synthetic ester fluid.

1.3 Classification. The fluorosilicone elastomer rubber should be of the following types, classes, and grades, as specified (see 6.2).

Type I - O-rings and other compression seals.
- Class 1 - O-rings (available only in Grades 60, 70, and 80)
- Class 3 O-rings (available only in Grade 75)

Type II - Molded parts (other than compression seals), sheets, strips, and extruded shapes.
- Class 1 - General purpose
  - Grade 40 - Nominal 40 type A durometer hardness, Type II only.
  - Grade 50 - Nominal 50 type A durometer hardness, Type II only.
  - Grade 60 - Nominal 60 type A durometer hardness, Types I and II.
  - Grade 70 - Nominal 70 type A durometer hardness, Types I and II.
  - Grade 80 - Nominal 80 type A durometer hardness, Types I and II.
- Class 2 - High-strength general purpose (available only in Grade 50), Type II only.
- Class 3 - High modulus, increased temperature resistance (available only in Grade 75), Types I and II.

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2. APPLICABLE DOCUMENTS

2.1 General**.** The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-R-25988/1	-	Rubber, Fluorosilicone Elastomer, Oil-and Fuel-Resistant, O-Rings, Class 1, Grade 70
MIL-R-25988/2	-	Rubber, Fluorosilicone Elastomer, Oil-and Fuel-Resistant, O-Rings, Class 3
MIL-R-25988/3	-	Rubber, Fluorosilicone, Elastomer, Oil-and Fuel-Resistant, O-Rings, Class 1, Grade 60
MIL-R-25988/4	-	Rubber, Fluorosilicone Elastomer, Oil-and Fuel-Resistant, O-Rings, Class 1, Grade 80

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-190	-	Identification Marking of Rubber Products
MIL-STD-289	-	Visual Inspection Guide for Rubber Sheet Material
MIL-STD-298	-	Visual Inspection Guide for Rubber Extruded Goods
MIL-STD-407	-	Visual Inspection Guide for Rubber Molded Items

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D297	-	Standard Test Methods for Rubber Products—Chemical Analysis
ASTM D395	-	Standard Test Methods for Rubber Property—Compression Set
ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension

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ASTM D471	-	Standard Test Method for Rubber Property—Effect of Liquids
ASTM D573	-	Standard Test Method for Rubber—Deterioration in an Air Oven
ASTM D624	-	Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers
ASTM D1329	-	Standard Test Method for Evaluating Rubber Property—Retraction At Lower Temperatures (TR Test)
ASTM D1414	-	Standard Test Methods for Rubber O-Rings
ASTM D2240	-	Standard Test Method for Rubber Property—Durometer Hardness

(Copies of these documents are available online at https://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories
ISO 3601-1	-	Fluid Power Systems- O-Rings – Part 1: Inside Diameters, Cross Sections, Tolerances and Designation Codes
ISO 3601-3	-	Fluid Power Systems- O-rings – Part 3: Quality Acceptance Criteria

(Copies of this document are available online at https://www.iso.org.)

SAE INTERNATIONAL

AMS2629	-	Fluid, Jet Reference
AMS3021	-	Fluid, Reference for Testing Di-Ester (Polyol) Resistant Material
AMS3400	-	Fluid, Reference for Testing Polyalpahaolefin (PAO) Resistant Material
ARP3050	-	Suitable Test Sizes for O-ring Specifications
AS568	-	Aerospace Size Standard for O-Rings
AS5752	-	Aerospace – Visual Inspection Standard for Elastomeric Sealing Elements Other than O-Rings
AS6414	-	Manufacturing Processing Requirements for Molded Elastomer Components Used in Aerospace Applications
AS9100	-	Quality Management Systems – Requirements for Aviation, Space and Defense Organizations
AS9120	-	Quality Management Systems – Requirements for Aviation, Space and Defense Distributors

(Copies of these documents are available online at https://www.sae.org)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Type I compression seals furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Qualification testing shall be performed (see 6.3) in accordance with 4.2 for all Type I compression seals.

3.2.1 Test report. A test report shall be prepared for each Type I class and grade. An updated test report shall be prepared whenever a change is made in the materials, processes, or this specification. Qualification testing for Type I shall be performed every 3 years. Supplier shall make qualification test report available upon request.

3.3 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3.1 Test report. A test report shall be prepared for each Type II class and grade. An updated test report shall be prepared whenever a change is made in the materials, processes, or this specification. First article tests over three years old shall not be acceptable. Supplier shall make tests reports available upon request.

3.4 Materials. The cured compound of fluorosilicone elastomer rubber shall be entirely suitable for the intended purpose and shall be of the same composition and quality as the materials used for the first article sample. Material shall be prepared from ingredients as shall be necessary to achieve the requirements detailed in this specification and shall be a compound based on a fluorosilicone elastomer, suitably cured to produce product meeting the requirements of 3.6. Material used shall be based on 100% virgin fluorosilicone elastomer. No reprocessed or non-fluorosilicone polymer is acceptable.

3.5 Dimensions and tolerances.

3.5.1 Sheets and strips. Unless otherwise specified (see 6.2), sheets and strips shall be supplied in the lengths and widths and to the length and width tolerances specified in 3.5.1.1 and 3.5.1.2, and the tolerance on thickness shall be compliant with tolerances shown in table I. The dimensions and tolerances of shapes cut from sheet and strip shall be as specified on the detail drawings.

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3.5.1.1 Sheets. The width of sheet material shall be 36 inches ± 1 inch (91.4 cm ± 2.54 cm), and the length shall be 120 inches (3.05 m) ± 1 percent.

3.5.1.2 Strips. The width of strip shall be 4 inches (102 mm) ± 5 percent and the length shall be 75 feet ± 1 foot (22.86 m ± 0.305 m).

TABLE I. Thickness tolerances of sheet and strip.

Nominal thickness		Tolerances	
(inch)	(mm)	(inch)	(mm)
0.031 and less	0.787	±0.010	±0.254
Over 0.031 to 0.063 inclusive	0.787 to 1.600	±0.012	±0.305
Over 0.063 to 0.125 inclusive	1.600 to 3.180	±0.016	±0.406
Over 0.125 to 0.188 inclusive	3.180 to 4.780	±0.020	±0.508
Over 0.188 to 0.375 inclusive	4.780 to 9.530	±0.031	±0.787
Over 0.375 to 0.563 inclusive	9.530 to 14.30	±0.047	±1.190
Over 0.563 to 0.750 inclusive	14.30 to 19.10	±0.063	±1.600
Over 0.750 to 1.000 inclusive	19.10 to 25.40	±0.093	±2.360
Over 1.000	25.40	±10%	±10%

3.5.2 Molded parts (excluding O-rings) and extruded shapes (including tubing). Unless otherwise specified (see 6.2), dimensions and tolerances of molded parts (excluding O-rings) and extruded shapes (including tubing) shall be as specified on the drawing.

3.5.3 O-rings. Dimensions and tolerances of O-rings shall be in accordance with the applicable military specification sheet as shown below. If not specified, O-rings standard sizes and tolerances are shown in AS568. The procedures outlined in Annex B of ISO 3601-1 shall be followed for dimensional inspection.

a. Class 1, Grade 60 MIL-R-25988/3

b. Class 1, Grade 70 MIL-R-25988/1

c. Class 1, Grade 80 MIL-R-25988/4

d. Class 3 MIL-R-25988/2

3.5.4 Color. Unless otherwise specified, material procured to this specification shall be blue. A pigment that can be used is Ferro Blue No. 3247 or equal (see 6.5). A formula of 1.5 parts of this pigment per 100.0 parts of fluorosilicone is suggested.

3.6 Physical properties. The physical properties of the fluorosilicone elastomer rubber shall conform to the values given in table II and table III.

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TABLE II. Physical properties, Type I (O-rings).

	Class 1 Grade 60	Class 1 Grade 70	Class 1 Grade 80	Class 3 Grade 75	Test Method
Original Properties					
Specific gravity	As determined	As determined	As determined	As determined	ASTM D1414 (D297) Hydrostatic Method
Hardness, Durometer Type A	60 ± 5	70 ± 5	80 ± 5	75 ± 5	ASTM D1414 (D2240)
Hardness, Durometer Type M	As determined	As determined	As determined	As determined700	
Tensile strength, psi, min.	700 (4.83 kPa)	750 (5.17 kPa)	750 (5.17 kPa)	(4.83 kPa)	ASTM D1414
Elongation, %, min.	150	125	70	90	ASTM D1414
Temperature retraction, ˚F, max.	-70 (-56.7 ˚C)	-70 (-56.7 ˚C)	-70 (-56.7 ˚C)	-70 (-56.7 ˚C)	Paragraph 4.6.1
After air aging 70 hours @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C)					Paragraph 4.6.5
Compression set, %, max.	20	15	25	25	ASTM D1414 (D395, Method B)
After air aging 70 hours @ the temperatures indicated:	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	437 ˚F ± 5 ˚F (225 ˚C ± 2.8 ˚C)	
Type A Hardness change, point, max.	+5, -5	+5, -5	+5, -5	+5, -5	
Type M Hardness change, point, max	+5, -5	+5, -5	+5, -5	+5, -5	ASTM D1414 (D573)
Tensile strength decrease, %, max.	20	15	15	35	
Elongation decrease, %, max.	20	20	20	45	Paragraph 4.6.2
Weight loss, %, max.	2	2	2	2	
After air aging 22 hours @ 347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C)					ASTM D1414 (D395, Method B)
Compression set, %, max.	25	10	20	25	Paragraph 4.6.5

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TABLE II. Physical properties, Type I (O-rings) – Continued.

	Class 1 Grade 60	Class 1 Grade 70	Class 1 Grade 80	Class 3 Grade 75	Test Method
After aging 22 hours @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C) in AMS2629, Type I:					
Type A Hardness change, point, max.	-20	-20	-20	-15	ASTM D1414 (D471)
Type M Hardness change, point, max.	-20	-20	-20	-20	
Tensile strength decrease, %, max.	30	25	25	30	Paragraph 4.6.4
Elongation decrease, %, max.	25	15	15	20	
Volume change, %, max.	1 to 15	1 to 15	1 to 15	1 to 15	Paragraph 4.6.5
Compression Set, % max	20	20	20	20	
After aging 70 hours in AMS3400 @ the temperatures indicated:	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	275 ˚F ± 5 ˚F (135 ˚C ± 2.8 ˚C)	
Type A Hardness change, point, max.	±15	±15	±15	±20	ASTM D1414 (D471)
Type M Hardness change, point, max	±15	±15	±15	±20	
Tensile strength decrease, %, max.	20	20	20	30	
Elongation decrease, %, max.	10	10	10	20	
Volume change, %, max.	1 to 10	1 to 10	1 to 10	1 to 15	
After aging 336 hours in AMS3400 @ 250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)					ASTM D1414 (D395, Method B)
Compression set, %, max.	25	15	15	25	
After aging 70 hours in AMS3021 @ the temperatures indicated:	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C)	Paragraph 4.6.3
Type A Hardness change, point, max.	±15	±15	±15	±20	
Type M Hardness change, point, max.	±15	±15	±15	±20	
Tensile strength decrease, %, max.	30	40	30	30	
Elongation decrease, %, max.	30	25	15	20	ASTM D1414 (D471)
Volume change, %, max.	1 to 15	1 to 15	1 to 15	1 to 15	
Compression set, %, max.	50	30	65	85	Paragraph 4.6.5

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TABLE III. Physical properties, Type II [molded parts (other than O-rings), sheets, strips, and extruded shapes].

	Class 1					Class 2	Class 3	
	Grade 40	**Grade 50**	**Grade 60**	**Grade 70**	**Grade 80**	**Grade 50**	**Grade 75**	**Test Method**
Original Properties								
Specific gravity	As determined	As determined	As determined	As determined	As determined	As determined	As determined	ASTM D297
Hardness, Durometer, Type A	40 ± 5	50 ± 5	60 ± 5	70 ± 5	80 ± 5	50 ± 5	75 ± 5	ASTM D2240
Tensile strength, psi, min	800 (5.52 kPa)	900 (6.20 kPa)	900 (6.20 kPa)	950 (6.55 kPa)	750 (5.17 kPa)	1150 (7.93 kPa)	800 (5.52 kPa)	ASTM D412
Elongation, %, min.	225	200	175	150	70	350	100	ASTM D412
Tensile stress @ 100% elongation, psi, min	—	—	—	—	—	—	700	
Tear strength, ppi, min	40 (276 Pa)	40 (276 Pa)	40 (276 Pa)	75 (517Pa)	40 (276 Pa)	150 (1034 Pa)	40 (276 Pa)	ASTM D624
Temperature retraction, ˚F, max.	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	-70 (-56.6 ˚C)	Paragraph 4.6.1
After air aging 70 hr @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C):								ASTM D395, Method B
Compression set, %, max.	15	15	15	15	15	15	15	Paragraph 4.6.5
After air aging 70 hr @ the temperatures indicated:	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	392 ˚F ± 5 ˚F (200 ˚C ± 2.8 ˚C)	437 ˚F ± 5 ˚F (225 ˚C ± 2.8 ˚C)	ASTM D573
Type A Hardness change, points, max.	+10, -5	+10, -5	+10, -5	+10, -5	+10, -5	+10, -5	+10, -5	
Tensile strength decrease, %, max.	30	25	25	20	20	30	35	
Elongation decrease, %, max.	25	25	25	25	20	30	45	
Weight loss, %, max.	2	2	2	2	2	2	2	Paragraph 4.6.2

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TABLE III. Physical properties, Type II [molded parts (other than O-rings), sheets, strips, and extruded shapes] – Continued

	Class 1					Class 2	Class 3	
	Grade 40	**Grade 50**	**Grade 60**	**Grade 70**	**Grade 80**	**Grade 50**	**Grade 75**	**Test Method**
After air aging 22 hrs. @ 347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C):Compression set, %, max.	25	35	40	25	45	40	35	ASTM D395, Method B Paragraph 4.6.5
After aging 70 hr in AMS3400 @ the temperatures indicated:	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	250 ˚F ± 5 ˚F (121 ˚C ± 2.8 ˚C)	347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C)	ASTM D471 Paragraph 4.6.3 Paragraph 4.6.5
Type A Hardness change, points, max	±15	±15	±15	±15	±15	±15	±20	
Tensile strength decrease, %, max.	45	45	45	40	30	40	30	
Elongation decrease, %, max.	30	30	30	25	15	30	20	
Volume Change, %	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	
Compression set, %, max.	35	35	45	30	60	45	80	
After aging 22 hr @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C) in AMS2629, Type 1:								ASTM D471 Paragraph 4.6.4 Paragraph 4.6.5
Type A Hardness change, points, max.	-15	-15	-20	-20	-20	-25	-15	
Tensile strength decrease, %, max.	65	55	50	45	30	40	35	
Elongation decrease, %, max.	60	50	40	35	30	30	30	
Volume change, %	1 to 30	1 to 25	1 to 25	1 to 25	1 to 25	1 to 25	1 to 25	
Compression set, %, max.	20	20	20	20	20	20	20	

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TABLE III. Physical properties, Type II [molded parts (other than O-rings), sheets, strips, and extruded shapes] – Continued

	Class 1					Class 2	Class 3	
	Grade 40	**Grade 50**	**Grade 60**	**Grade 70**	**Grade 80**	**Grade 50**	**Grade 75**	**Test Method**
After aging 70 hr in AMS3021 @ the temperatures indicated:	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	302 ˚F ± 5 ˚F (150 ˚C ± 2.8 ˚C)	347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C)	Paragraph 4.6.3
Type A Hardness change, points, max	±15	±15	±15	±15	±15	±15	±20	
Tensile strength decrease, %, max.	45	45	45	40	30	40	30	ASTM D471
Elongation decrease, %, max.	30	30	30	25	15	30	20	
Volume Change, %	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	1 to 15	
Compression set, %, max.	35	35	45	30	60	45	80	Paragraph 4.6.5

3.7 Identification of product.

3.7.1 Sheet and strip. Unless otherwise specified (see 6.2), sheet material (including strip cut from sheet) shall be marked to show the specification number; the manufacturer; the manufacturer's designation (compound number); type, class, and grade (if applicable); and the date of manufacture by quarter and year (for example, 1Q82). The following is an example of how the sheet material should be marked:

MIL-DTL-25988D XYZ Co.
Compound No.
Type II, Class 1, Grade 60
Date of manufacture

The material shall be marked for identification in compliance with MIL-STD-190. The identification shall be applied in recurring pattern in lines from one end of the sheet to the other and placed in intervals spaced 8 inches (203.2 mm), maximum, apart on the line

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and positioned in rows 5 inches ± 0.5-inch (127 mm ± 12.7 mm) apart lengthwise of the sheet. The marking shall be applied by suitable means using a marking fluid that is not deleterious to the fluorosilicone elastomer rubber. The marking shall not be obliterated by normal handling, and the color shall be of sufficient contrast to permit easy reading.

3.7.2 Molded parts and extruded shapes (including tubing). Where the size of the product permits, the identification shall be marked as indicated in 3.7.1. When identification marking of the product is impractical, the unit package shall show the compound number and manufacturer in addition to those markings specified in 6.2. Temporary marking of Type I parts for the manufacturer's identification shall be permitted at the time of manufacture but shall in no way affect the properties or function of the compression seal.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification (see 4.2).

b. First article inspection (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification.

4.2.1 Samples. Test samples for Type I material shall be restricted to O-rings from a production lot of elastomer conforming to size -214 of MIL-R-25988/1. Hardness shall be determined on specimens of sufficient dimensions to comply with ASTM D2240.

4.2.2 Tests. Qualification tests shall consist of all the tests specified in table II and paragraph 4.5 and shall be performed at a laboratory accredited to ISO/IEC 17025. All technical requirements shall be performed prior to the initial shipment of the product to the purchaser. Any change in the process or ingredients that would require requalification are listed in the SAE AS6414 document unless otherwise specified by the purchase order, print, or design data.

4.2.3 Qualification test report. Qualification tests are required for each Type I class and grade of material furnished to this specification. After the supplier completes the qualification testing, he shall prepare a test report (see 6.3.1).

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4.2.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or applicable paragraphs in the specification.

4.3 First article inspection.

4.3.1 Samples. Samples for Type II materials shall be obtained from 6 inches by 6 inches by 0.075-inch (152.4 mm by 152.4 mm by 1.905 mm) platen sheets. All samples shall be of identical composition and comparable state of cure as that of the end items.

4.3.2 Tests. First article tests shall consist of all the tests specified in table III and paragraph 4.5 and shall be performed at a laboratory accredited to AS9100 or ISO/IEC 17025.

4.3.3 First article test report. First article tests are required for each Type II, class, and grade of material furnished to this specification. After the supplier completes the first article inspection, he shall prepare a first article test report (see 6.4).

4.3.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or the applicable paragraphs in the specification.

4.4 Conformance Inspection.

4.4.1 Sampling for inspection. The method shall be specified in a purchase order/contract. If not specified, product shall be taken at random from each lot to perform all the required acceptance tests. The number of test iterations for each requirement shall be specified in the applicable test procedure.

4.4.1.1 Batch. Batch and the blending of compound are defined in AS6414.

4.4.1.2 Lot. Lot and Lot testing are defined in AS6414.

4.4.2 Conformance test samples. Whenever possible, the end item, or specimens cut from the end item, shall be used as the sample. If these items are unsuitable for use as test samples, tests shall be performed on samples of identical composition and comparable state of cure as that of the end item. O-ring sizes that are suitable for testing (tensile and elongation) are shown in ARP3050.

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4.4.3 Inspection of materials and components. The supplier is responsible for ensuring that materials and components used were manufactured, tested, and inspected in accordance with referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.4.4 Inspection of the end item.

4.4.4.1 Type I compression seals. End item inspection shall be as specified in the parts standard, drawing, or purchase document. If not specified, it shall be as follows: Prior to inspection, mold flash shall be removed from the parts in such a manner that they conform to the requirements specified herein. Each individual part (100%) shall be visually inspected according to ISO 3601-3, Grade -S or AS5752 as applicable, using a minimum of 1X magnification. The entire surface of the part shall be manually inspected. Dimensional tolerances shall be as specified in the applicable slash sheet.

4.4.4.2 Type II. Product shall be inspected according to MIL-STD-407 under 1X magnification minimum. The sample unit shall be 1 linear yard (91.4 mm), except if the end item is less than 1 yard, then the sample unit shall be the end item. The examination shall be in accordance with MIL-STD-289 and MIL-STD-298, as applicable. Defects in marking such as “incomplete, not legibly identified” or not as specified in 3.7 shall be considered minor. The sample size shall be in accordance with ANSI/ASQ Z1.4 single sampling plan inspection Level II with an AQL 1.0 except that the acceptance number shall be zero. The dimensions shall be within the tolerances specified in 3.5. Dimensions for extruded shapes shall be as specified on the drawing or in the contract or order.

4.5 Conformance tests. The following tests shall be conducted on each batch and lot of material. The manufacturer of the product shall be responsible for performing all required tests. Purchaser reserves the right to sample and perform any testing deemed necessary to ensure that the product conforms to the MIL-DTL-25988 requirements.

4.5.1 Batch Testing

AS6414 defines batch testing. Testing on the same batch and state of cure are acceptable as long as the AS568-214 O-rings are cured and tested within ninety (90) days of the manufacture of the subject lot.

Batch and the blending of compound as defined in AS6414. All incoming batches and or blended batches shall be tested per table IV and table V.

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Table IV. Batch Test for Type I

	Class 1 Grade 60	Class 1 Grade 70	Class 1 Grade 80	Class 3 Grade 75	Test Method
Original Properties					
Specific gravity	±0.02 from qualification value	±0.02 from qualification value	±0.02 from qualification value	±0.02 from qualification value	ASTM D1414 (D297) Hydrostatic Method
Hardness, Durometer Type A	60 ± 5	70 ± 5	80 ± 5	75 ± 5	ASTM D1414 (D2240)
Tensile strength, psi, min.	700 (4.83 kPa)	750 (5.17 kPa)	750 (5.17 kPa)	700 (4.83 kPa)	ASTM D1414
Elongation, %, min.	150	125	70	90	ASTM D1414
After air aging 22 hours @ 347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C) Compression set, %, max.	25	10	20	25	Paragraph 4.6.5 ASTM D1414 (D395, Method B)
After aging 22 hours @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C) in AMS2629, Type I: Volume change, %, max.	1 to 15	1 to 15	1 to 15	1 to 15	Paragraph 4.6.4 ASTM D1414 (D471)

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Table V. Batch Test for Type II

	Class 1					Class 2	Class 3	
	Grade 40	**Grade 50**	**Grade 60**	**Grade 70**	**Grade 80**	**Grade 50**	**Grade 75**	**Test Method**
Specific gravity	±0.02 from First Article value	±0.02 from First Article value	±0.02 from First Article value	±0.02 from First Article value	±0.02 from First Article value	±0.02 from First Article value	±0.02 from First Article value	ASTM D297 Hydrostatic Method
Type A Hardness Change, points, max,	40 ± 5	50 ± 5	60 ± 5	70 ± 5	80 ± 5	50 ± 5	75 ± 5	ASTM D1414 D2240
Tensile strength, psi, min	800 (5.52 kPa)	900 (6.20 kPa)	900 (6.20 kPa)	950 (6.55 kPa)	750 (5.17 kPa)	1150 (7.93 kPa)	800 (5.52 kPa)	ASTM D412
Elongation, %, min.	225	200	175	150	70	350	100	ASTM D412
After air aging 22 hr @ 347 ˚F ± 5 ˚F (175 ˚C ±2.8 ˚C): Compression set, %, max.	25	35	40	25	45	40	35	Paragraph 4.6.5 ASTM D395, Method B
After aging 22 hr @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C) in AMS2629: Volume change, %	1 to 30	1 to 25	1 to 25	1 to 25	1 to 25	1 to 25	1 to 25	Paragraph 4.6.4 ASTM D471

4.5.2 Lot Testing

Lot Tests for O-rings: Requirements shown in table VI are lot tests and shall be performed on each lot on the O-rings being supplied. It shall be permitted to use multiple such O-rings which shall be made from the same lot and batch to perform the applicable test. The test specimen for lot testing shall be the O-ring supplied unless the supplied O-ring size is such that the test cannot be performed on it. In that case, use of MIL-R-25988/1-214 O-rings to perform that specific test shall be permitted. O-ring sizes that are suitable for tensile and elongation testing are shown in ARP3050.

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Table VI. Lot Test for Type I O-rings

	Class 1 Grade 60	Class 1 Grade 70	Class 1 Grade 80	Class 3 Grade 75	Test Method
Original Properties					
Specific gravity	±0.02 from qualification value	±0.02 from qualification value	±0.02 from qualification value	±0.02 from qualification value	ASTM D1414 (D297) Hydrostatic Method
Hardness, Durometer Type A	60 ± 5	70 ± 5	80 ± 5	75 ± 5	ASTM D1414 (ASTM D2240)
Tensile strength, psi, min.	700 (4.83 kPa)	750 (5.17 kPa)	750 (5.17 kPa)	700 (4.83 kPa)	ASTM D1414
Elongation, %, min.	150	125	70	90	ASTM D1414
After air aging 22 hr @ 347 ˚F ± 5 ˚F (175 ˚C ± 2.8 ˚C)					Paragraph 4.6.5
Compression set, %, max.	25	10	20	25	ASTM D1414 (D395, Method B)
After aging 22 hr @ 75 ˚F ± 5 ˚F (23.9 ˚C ± 2.8 ˚C) in AMS2629, Type I:					Paragraph 4.6.4
Volume change, %, max.	1 to 15	1 to 15	1 to 15	1 to 15	ASTM D1414 (D471)

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4.6 Test Methods

4.6.1 Temperature retraction (TR-10). The temperature retraction test shall be conducted in accordance with ASTM D1329. Three O-ring specimens as specified in 4.2.1 shall be used for Type I and shall be elongated 50 percent in accordance with the rod markings. The temperature retraction test for Type II shall be conducted on three 2-inch (50.8-mm) specimens. Elongation of Type II materials shall be 50 percent of the ultimate elongation. The test shall be conducted at a heating rate of 1 ˚C per minute.

4.6.2. Weight loss. The weight loss test shall be conducted on samples air aged in accordance with table II and table III. Test specimens shall be conditioned for 24 hours in a desiccators before and after air aging. The specimens shall be weighed immediately after the desiccation period before and after aging. The percentage weight loss shall be calculated as follows:

$$\frac{W_1 - W_2}{W_1} \times 100$$

W_1 = weight of sample before air aging.
W_2 = weight of sample after air aging.

4.6.3 Engine oil aging. Engine oil aging of specimens shall be conducted in clean 39 millimeters (mm) OD by 300 mm Pyrex glass test tubes fitted with two-hole cork stoppers. Each stopper shall be fitted with two lengths of 8 mm Pyrex glass tubing (chimneys), one 3 inches (76.2 mm) in length and the other 5 inches (127 mm) in length. The 3-inch (76.2 mm) chimney shall extend through and one-half (12.7 mm) inch above the top of the stopper. The 5-inch (127 mm) chimney shall extend through and 3½ inches (88.9 mm) above the top of the stopper. An aluminum block heater shall be used for aging the specimens. Use of an oil bath is permissible but is not desired. For each test, 140 milliliters (mL) of oil shall be used. The test tubes shall be inserted into the aluminum block or oil bath in such manner that the fluid level in the test tubes shall be approximately 1⅝ inches (41.3 mm) above the heating unit of the aluminum block or surface of the oil bath. This distance shall not be measured from the top of the aluminum block or covering of the oil bath. The specimens shall be suspended in the oil by soft iron or nichrome wire hangers as follows:

a. Type 1: Three specimens shall be suspended horizontally in the fluid, one each at depths of 1½ inches (38.1 mm), 3½ inches (88.9 mm), and 5½ inches (140 mm) below the surface of the fluid. Hardness specimens shall be placed in a separate chimney stoppered tube keeping approximately the same rubber to oil ratio.

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b. Type II: Four dumbbells cut with a die "C," conforming to ASTM D412, shall be tested. There shall be only two specimens aged in a single test tube. The specimens shall be suspended vertically in the fluid, one each at depths of 1-inch (25.4 mm) and 2 inches (50.8 mm) below the surface of the fluid. The measurement of specimen depth in the fluid shall be made between the top edge of the specimen and the fluid level. Care shall be taken to prevent contact of specimens with each other or with the wall of the test tube.

The oil aging time and temperature shall be in accordance with table II. After aging and prior to the physical property determinations, the specimens shall be removed from the hot fluid and cooled 30 minutes in fresh fluid. Tensile strength, elongation, hardness, volume change and compression set shall then be determined as specified in table II.

4.6.4 Fuel aging. Specimens shall be immersed in test fluid conforming to AMS2629, Type 1 for 22 hours at room temperature. The ratio of fluid to rubber shall be approximately 40 ml per gram. Aging shall be conducted in a glass container. An airtight seal shall be maintained for the duration of the test. One sample at a time shall be removed from the fluid, immediately blotted dry, and tested.

4.6.5 Compression set. Except where otherwise specified herein, compression set shall be determined in accordance with ASTM D395, Method B. Specimens for Type I materials shall be two O-rings. Two circular plied-up buttons with dimensions of 1.129 ± 0.0001 inches (28.677 ± 0.25 mm) diameter and approximately 0.5-inch (12.7 mm) thick shall be cut from sheet for type II material.

4.6.5.1 Engine oil aged**.** Compression set shall be determined on specimens aged 70 hours at the applicable temperature as specified in table II, immersed in AMS 3021. The compression set plates for testing Type I material shall be approximately 0.375 inches by 2 inches by 4 inches (9.52 mm by 50.8 mm by 102 mm). There shall be six, 1/4 inches (6.35 mm) bolt holes, one on each corner and one located in the middle of each 4-inch (102 mm) edge and on the center line of the corner holes. There shall also be ¼-inch (6.35 mm) holes through the middle of each half of the plates to allow fluid to be in contact with the inside diameter of the O-rings. The compression set plates for Type II material shall be in accordance with ASTM D395. The original thickness of the specimens shall be measured and the test fixtures shall be assembled using two test specimens. The specimens shall be compressed 25 percent. The test fixture shall be placed in a 1 liter stainless beaker. The beaker shall be fitted with a suitable vented stainless steel cap. The cap shall be sealed with an O-ring conforming to Type I, Class I of this specification (size –240, MIL-DTL-25988/1, has been used). The beaker shall be placed in a suitable oven at the applicable temperature with vent open. After the fluid has reached the test temperature (approximately two hours), the vent shall be closed and the beaker shall be left in the oven for a total aging time of 70 hours. At the end of the aging time, the specimens shall be removed from

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the compression plates immediately and allowed to cool on paper towels for 30 minutes. The excess fluid shall be blotted from the specimens and the final thickness measurement made.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency or within the military service's system commands. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rubber covered by this specification is for use where resistance to jet fuel and lubricating oil is required. The elevated test temperatures (see table II) were selected as the maximum temperatures at which good reproducible laboratory data could be obtained. These temperatures do not necessarily indicate nor are they to be construed as the maximum operating temperatures at which these materials can be used.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and grade (see 1.3).

c. First article when required (see 4.3).

d. Dimensions and tolerances (see 3.5).

e. Quantity.

f. Detail drawing and additional requirements, if any.

g. Packaging requirements (see 5.1).

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 25988 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from (insert name, mailing address, and email of qualifying activity). An online listing of products qualified to this specification may be found in Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification Testing. Qualification is required for all Type I materials. Testing will be as specified in 3.2 and 4.2. Requalification is required every 3 years or as directed by AFRL/RXSS.

A copy of the qualification test report along with the laboratory certification shall be furnished to the Air Force Research Laboratory, ATTN: AFRL/RXSS, 2179 12th Street, Wright-Patterson AFB, OH 45433-7718. Test results to previous revisions of this specification are not acceptable.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample. Testing will be as specified in 3.3 and 4.3. The first article should consist of four units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the first article.

A copy of the first article test report should be furnished to the Air Force Research Laboratory, ATTN: AFRL/RXSS, 2179 12th Street, Wright-Patterson AFB, OH 45433-7718. The first article test does not need to be repeated for new orders or different parts provided the materials and processes have not been changed and a certified statement to this effect is furnished to the procuring activity. The waiving of the first article tests will be strictly at the discretion of the procuring activity. Test results to previous revisions of this specification are not acceptable.

6.5 Source for color pigment. A source for the color pigment specified in 3.5.4 is Ferro Corporation, 4150 East 56th St., Cleveland, Ohio 44105.

6.6 Subject term (key word) listing.

Aging
Compression
Desiccation
Elongation
Fluorosilicone
Hardness
Hydrostatic
O-rings

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6.7 Summary of changes. This revision updated military and non-government references in section 2, revised certain requirements in table II and table III, updated quality assurance provisions in section 6 and added a qualification requirement for Type I compression seals.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – AS
Air Force – 20
DLA – IS

Preparing activity:
Air Force – 20

Review activities:
Army – MI
Navy – OS
Air Force – 71

Agent:
Air Force - 110

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